# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building server-side applications in **Java** (Spring Boot, Maven) and Python, with **multithreading**, object-oriented design patterns, SQL and NoSQL databases, **AWS** and Azure, and Jenkins CI/CD in Agile teams.

## Education

**University of Southern California, Los Angeles, CA** — **August 2024 – May 2026**
*Master of Science in Computer Science* — *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

**Pandit Deendayal Energy University, Gandhinagar, India** — **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* — *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Operating Systems, Software Engineering, Data Retrieval

## Experience

**USC Dornsife** — **September 2025 – Present**
*Software Engineer* — *Los Angeles, CA*

- Designed normalized **SQL** schemas on Azure SQL, SQL Server, and PostgreSQL with row-level security for 500+ users, cutting dashboard latency from 1.2s to 350ms through indexing and query tuning.
- Built a **distributed** async job system with background workers, explicit state transitions, and failure isolation, processing 1K+ jobs at 3x throughput and sustaining 99%+ reliability for 5K+ monthly integrations.

**Grad8** — **June 2025 – August 2025**
*Software Engineer Intern* — *Los Angeles, CA*

- Built a **multithreaded** batch-processing engine with a thread pool executor for PDF, DOCX, and ZIP ingestion, processing 1K+ files per batch with fault-tolerant execution behind REST APIs.
- Developed server-side REST APIs with modular MVC structure on MySQL and AWS RDS, serving 150+ instructors and 10K+ daily requests with stable latency, shipped through **CI/CD** pipelines.

**Aasma Technology Solutions** — **August 2023 – July 2024**
*Software Developer Engineer – SecureSync* — *India*

- Built object-oriented modules using **Singleton and Factory** design patterns for large-scale API integrations, improving response times 30% and reducing technical debt 20% across a cross-platform microservice data-sharing platform.
- Wrote 20+ unit and integration test suites with **Mockito** and PyTest, executed nightly by Jenkins across Linux, Windows, and macOS with test reports, in Agile sprints with Microsoft and GitHub.

**Aasma Technology Solutions** — **May 2023 – August 2023**
*Full Stack Developer Intern – VibrantChem* — *India*

- Built eight **Java Spring Boot** microservices with Maven for authentication, resource allocation, billing, and notifications on MongoDB and Azure SQL, synchronizing 2M+ records across locations in real time.
- Secured services with JWT and role-based access, ran Snyk and OWASP **security scanning** in Jenkins CI/CD with unit, integration, and end-to-end tests, and reached 95%+ build validation.

**LST Capitals** — **November 2022 – January 2023**
*Web Developer Intern* — *India*

- Developed **Java Spring Boot** REST APIs for bookings, pricing, coupons, and payments with transactional locking to prevent double-booked time slots, sustaining sub-200ms latency across 50K+ booking records under concurrent use.
- Gathered requirements from the client and mapped 12+ **integration** points into an end-to-end architecture, then automated refund and dispute processing with idempotent webhooks, cutting manual work 70%.

## Skills

**Languages:** Java, Python, SQL, JavaScript, TypeScript, C++, Shell
**Java & Backend:** Spring Boot, Maven, Java 8 Streams and Lambdas, Multithreading, REST APIs, Microservices, Mockito
**Databases:** PostgreSQL, SQL Server, MySQL, Oracle, MongoDB, Schema Design, Indexing
**Cloud & DevOps:** AWS (EC2, S3, RDS), Microsoft Azure, Docker, Jenkins, Git, GitHub Actions, CI/CD, Linux
**Security & Networking:** JWT, RBAC, Snyk, OWASP, TCP/IP, HTTP
**Fundamentals:** Data Structures, Algorithms, OOP, Design Patterns, Distributed Systems, Agile/Scrum

## Projects

**Chat2SQL** Code | *Java 17, Maven, TypeScript, Oracle, MySQL, Docker*

- Extended a large **Java** 17 and TypeScript database workspace with Maven and Docker builds, supporting 10+ databases including Oracle, MySQL, and MongoDB and generating schema-aware SQL at 82% accuracy.

**Concurrent Order Processing Service** | *Java 17, Spring Boot, PostgreSQL, Kafka, JUnit, Docker*

- Designed a **Spring Boot** service that processes orders with a thread pool and CompletableFuture pipelines, uses Java Streams for aggregation, publishes events to Kafka, and is covered by JUnit tests.

**CPU-Scheduling Simulator** Live | *C++, React.js, JavaScript, EJ-2*

- Implemented FCFS, SJF, SRT, Round Robin, priority, and multilevel-queue **scheduling algorithms** in C++, testing correctness across 1K+ process combinations and showing up to 25% lower average turnaround time.